

Surya Vikram Singh

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SUMMARY

Software engineer with 4+ years building and operating production backend services, now on machine learning systems: model serving, evaluation, and LLM and agent applications. I design for reliability and scaling and own the telemetry and tracing underneath. Full lifecycle: architecture, code review, testing, CI/CD, on-call, and integrating into partner codebases.

EXPERIENCE

Shivoy, Inc | Applied AI Engineer

May 2025 – Present | Virginia, USA

- Designed and shipped a centralized MCP tool server with semantic tool discovery that loads only the 2-3 query-relevant schemas per turn instead of the full catalog, cutting input tokens by over 50% and inference cost along with them.
- Re-architected the agent execution loop from iterative tool-calling to code-execution-in-sandbox, letting the model chain tools, transform outputs, and persist state across turns, which removed repeated serialization and cut round-trips per task.
- Built the retrieval service behind the assistant: hybrid dense-vector plus BM25 search on OpenSearch with Titan embeddings and reranking over top-k, tuned on retrieval-quality metrics and served through FastAPI endpoints.
- Owned testing and release quality by building an LLM-as-a-judge evaluation harness that gates prompt, embedding, and retrieval changes, blocking regressions from reaching production until they clear a curated eval set.
- Debugged production issues across the stack – retrieval misses, tool-call failures, latency regressions – using distributed traces and reproducible test cases, and partnered with product and science stakeholders on requirements and rollout.

CSUF, Office of Graduate Studies | Software Engineer, AI/ML

Oct 2024 – May 2025 | Fullerton, CA

- Built an internal analytics agent on LangChain's Pandas DataFrame Agent behind a FastAPI backend on AWS Lambda and API Gateway, turning natural-language questions from non-technical staff into validated queries and structured responses.
- Added a Python REPL tool generating Plotly visualizations on the fly, delivered through a Streamlit interface, cutting reporting effort by 70% while iterating weekly with the staff who actually used it.

iNeuron | Software Engineer

Aug 2020 – Jul 2023 | Bengaluru, India

- Designed and owned the Flask backend serving layer for a production recommendation system, defining the service contract that decoupled API consumers from the model layer so either side could change independently.
- Built the content-based recommendation engine scoring similarity across text, skill tags, and difficulty with TF-IDF and cosine similarity, improving accuracy 25% through metadata cleaning and feature engineering.
- Scaled the read path with a precompute-and-cache architecture – batch-generated top-N results in Redis with TTL invalidation – removing per-request model computation and cutting p95 latency from 220ms to under 40ms at 500 req/sec.
- Containerized the services with Docker and deployed on AWS (Lambda, API Gateway, ECS/Fargate) with CloudWatch metrics and alarms on latency and error rate routed to on-call, and participated in code reviews and design discussions.

PROJECTS

Multi-Agent Research Assistant ([GitHub](#))

- Built a distributed multi-agent system on LangGraph with a supervisor that decomposes each request, fans out to specialist agents running concurrently against independent data sources, then merges and de-duplicates their results.

Open-Source Model Fine-Tuning & Quantization ([HuggingFace](#))

- Fine-tuned Qwen3 4B and LFM2.5 1.2B on a domain-specific DSL and Qwen3-TTS 1.7B on LJSpeech for single-speaker voice adaptation; published MLX 4/6/8-bit and GGUF builds plus a 4-bit multimodal MedGemma 4B, all running on consumer hardware.

SKILLS

Languages: Python, SQL, TypeScript, Go, Bash, Git; object-oriented design, design patterns, code review, testing

Distributed Systems: Service design and architecture, microservices, event-driven pipelines, caching and invalidation, fault tolerance and graceful degradation, latency and throughput tuning, reliability and scaling

Observability & Operations: Telemetry and metrics collection, monitoring and alarms, distributed tracing, structured logging, health checks, alerting and on-call escalation, latency and error-rate dashboards, CloudWatch, performance profiling

ML & Generative AI: Model serving and deployment, PyTorch, PyTorch Lightning, SageMaker, Bedrock, Hugging Face, fine-tuning (LoRA/QLoRA), quantization, LLM and agent applications (LangChain, LangGraph, MCP), RAG, evaluation

Backend & Data: FastAPI, Flask, REST APIs, Pydantic, async/await, background workers, PostgreSQL, DynamoDB, Redis, S3, OpenSearch, schema validation, batch and streaming ingestion, ETL and event-driven pipelines

Cloud & SDLC: AWS (EC2, Lambda, S3, ECS/Fargate, API Gateway, CloudWatch, SageMaker, Bedrock), Docker, Kubernetes, CI/CD (GitHub Actions), build and release automation, source control, documentation, on-call

EDUCATION

Master of Science in Information Systems | Data Science & ML Specialization | Cal State University, Fullerton Aug 2023 – May 2025